

Parent Guide — middle / module-14- magic-squares

IKS Curriculum

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Parent Guide — Module 14: Magic Squares (*Bhadragaṇita*)

Your child's age: Middle (Grades 6–8) · **Module length:** 10 sessions × 45 min

What your child is learning

A **magic square** is a square grid of distinct numbers where every row, every column, and both main diagonals add up to the same total — the *magic constant*. The smallest non-trivial example uses the numbers 1 to 9 in a 3×3 grid; every line sums to 15.

The construction method. Your child will learn an algorithm — Nārāyaṇa Paṇḍita's *turagagati* (“horse-step”) method — for building a magic square of any odd order (3×3 , 5×5 , 7×7 , ...). The same method appears in modern textbooks under the names *Siamese method* or *de la Loubère method*; Nārāyaṇa documented it nearly 300 years before the European description.

The history. Nārāyaṇa Paṇḍita wrote his *Gaṇita-kaumudī* in 1356 CE. Book 14 of that work, *Bhadragaṇita*, is the most systematic pre-modern treatment of magic squares we have in writing. Your child will also meet the Chinese *Lo Shu* square (~5th c. BCE earliest evidence), the Khajuraho temple inscription (Madhya Pradesh, ~10th–11th c. CE), and Albrecht Dürer's 1514 European square — to see that the topic appears across cultures.

A note on the cultural framing

Magic squares have been used in some traditional contexts as *yantras* — protective or contemplative diagrams. In this curriculum we study them as **mathematics**: as a problem in combinatorics. We acknowledge the broader cultural uses without engaging in any ritual framing. The mathematics is real; the cultural breadth is real; the curriculum keeps a clear scholarly stance.

What to expect this fortnight

Daily warm-up: 2 minutes of “row sum recall” — your child is shown three numbers from a known row and asked to give the fourth.

Hands-on construction: Your child will build several magic squares with their own hand, including a 5×5 from scratch. Bring blank graph paper home if they ask.

Project (Days 8–10): Each pair designs and presents an original magic square (5×5 or larger). Three format options: poster, short demo video, or written exposition with diagrams.

How to help

Helpful: - Let your child show you the *turagagati* algorithm. Watching them teach it is the best check on understanding. - Try the 3×3 yourself. Ask: “Why is 5 always in the centre?” - For the project, offer to be the test audience. Ask “If I didn’t know the method, would I follow your explanation?”

Not helpful: - Looking up a 5×5 magic square online and showing it to them. The point is the *construction*, not the final pattern. - Telling them it’s “ancient Indian magic.” It’s not magic. It’s a careful algorithm, and the framing matters for their relationship with mathematics.

What success looks like

By the end of the module, your child can: - State the magic constant formula $S = n(n^2 + 1)/2$ and apply it for $n = 3, 4, 5, 6, 7$. - Construct a 5×5 magic square from scratch using the stair-step method. - Verify that a given grid is or is not magic by summing rows, columns, and diagonals. - Place Nārāyaṇa Paṇḍita’s *Gaṇita-kaumudī* (1356 CE) in the chronology alongside Lo Shu and Dürer. - Explain why the Khajuraho 4×4 square is “extra magic” — its 2×2 panels also sum to 34.

Citation standard

Projects cite Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (1356 CE). The full citation chain is in the module’s `sources.md`.

Want to go deeper?

Senior band (Grade 9–12) generalises Nārāyaṇa’s method, proves *why* it produces a magic square, and explores symmetry classes of $n \times n$ magic squares. Preview: `curriculum/senior/module-14-magic-squares/README.md`.

10-Day Lesson Plan

Module 14 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	What is a magic square?	days/day-01-what-is-a-magic-	Discovery worksheet

Day	Session title	Lesson file	Activity / assessment
2	The magic constant	square.md days/day-02-magic-constant.md	Formative quiz Part A
3	How many 3×3 squares are there?	days/day-03-counting-and-symmetry.md	Pair worksheet
4	The stair-step method	days/day-04-stair-step-method.md	activities/activity-01-stair-step-drill.md
5	A 5×5 magic square	days/day-05-five-by-five.md	Formative quiz Part B
6	The 4×4 and Khajuraho	days/day-06-khajuraho.md	Class discussion
7	Nārāyaṇa and <i>Gaṇita-kaumudī</i>	days/day-07-narayana.md	Reflection paragraph
8	Lo Shu and Dürer	days/day-08-cross-cultural.md	Project planning + activities/activity-02-cross-cultural-compare.md
9	Sudoku + project session	days/day-09-sudoku-and-project.md	Project draft
10	Summative + share	days/day-10-summative-and-share.md	quizzes/summative.md + project

Pacing notes

- Days 1–2 are the conceptual core. Don’t rush the *discovery* of the 3×3 — most of the lasting motivation comes from a student building one themselves.
- Day 4 introduces the *turagagati* method on the 3×3 (which students already know). This lets the algorithm be the focus, not the construction.
- Day 5 applies the same algorithm to a 5×5. If the class struggled with Day 4, push Day 5 by one day and combine it with Day 6.
- Day 6 (4×4 Khajuraho) is the cultural-context highlight. It is NOT required that students build a 4×4 themselves at Middle band — that’s a Senior task. Showing the inscription and counting its panel-magic property is enough.
- Days 8–10 are project + presentation. Light enough to combine if school calendar requires.

Daily warm-up (recurring)

Every class starts with **2 minutes of “row sum recall”**: teacher writes any 3 numbers from a known magic square; students name the 4th number that completes the row to 15. (E.g. “1, 5, ?” → 9.) Builds arithmetic fluency.

What students leave the module with

- One personally constructed 5×5 magic square they can show a friend.
- The formula $S = n(n^2 + 1)/2$ memorised and explained.
- An understanding that magic squares appeared independently in multiple cultures — and that Nārāyaṇa Paṇḍita (1356 CE) produced the most systematic pre-modern treatment.
- A poster, demo, or short write-up as their project artefact.

Homework

Homework — Module 14 Magic Squares (Middle)

Day 1 → Day 2

Task: Copy the 3×3 magic square from class. Beside it, draw three rotations (90°, 180°, 270°). Verify one row of each.

Time: 10 min.

Day 2 → Day 3

Task: Verify the magic constant formula $S = n(n^2 + 1)/2$ for $n = 8$ and $n = 9$. Show your arithmetic. (Answers: 260 and 369.)

Time: 8 min.

Day 3 → Day 4

Task: Take the 3×3 magic square and add 10 to every entry. The square becomes one with numbers 11–19 instead of 1–9. What is the new magic constant? Verify one row.

Time: 10 min.

Day 4 → Day 5

Task: Construct three 5×5 magic squares from scratch using the *turagagati* method. Verify the magic constant of each (should be 65). Yes, they should all be the same square — that’s part of the discovery.

Time: 20 min.

Day 5 → Day 6

Task: Construct a 7×7 magic square using the stair-step method. Magic constant should be 175. Bring it tomorrow for spot-checks.

Time: 25 min.

Day 6 → Day 7

Task: Pick any 2×2 panel of the Khajuraho square (including one that wraps the edge). Verify the sum is 34. Write down both the panel cells and the sum.

Time: 5 min.

Day 7 → Day 8

Task: Polish your in-class reflection paragraph on Nārāyaṇa. Add one more specific example from your own construction work.

Time: 15 min.

Day 8 → Day 9

Task: Submit your project plan (4–6 sentences): format, target audience, the magic square you'll construct, what is unique about your design.

Time: 15 min.

Day 9 → Day 10

Task: Finalise project artefact and rehearse the 2-minute presentation aloud at least once. Time yourself.

Time: 30 min (cap).